# Roy (Yuanxi) Cheng

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## Education

**University of California, Los Angeles (UCLA)** Sep 2023 – Mar 2027

*Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering* *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Neural Networks & Deep Learning, Data Analysis & Regression, Data Mining, Probability & Statistical Inference, Databases, NLP, Monte Carlo, Optimization

## Projects and Research

**Biomedical Tissue Imaging** | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article evaluating CNN-based instance segmentation architectures with AP metrics, benchmarking cytoplasm boundary precision for biomedical imaging
- Converted 4k+ CytoNuke annotations to COCO format through a custom data pipeline, enabling standardized training across 3 segmentation architectures
- Trained and tuned YOLOv8 instance segmentation models in PyTorch, improving cytoplasm AP50 by 5% over the Cyto-RCNN baseline

**COVID Pandemic Effects on Secondary Education** | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned and merged 100k+ student records across multi-level datasets with pandas, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time series visualizations, revealing a 15% average decline in student performance
- Created an interactive Tableau map linking geographic data to score changes across 50 U.S. states for a public research website

**Adrenaline and Cognitive Fatigue Effects on Athletic Performance** | *R, ggplot2, tidyverse, dplyr*

- Designed $2^3$ factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Validated ANOVA assumptions using log transformation and ggplot2 diagnostics, confirming normality and homoscedasticity of residuals

## Work and Leadership Experience

**Capital One** McLean, Virginia

*Product Management and Analysis Intern* *Jun 2026 – Present*

- Analyze corporate-wide people data with SQL and Python, identifying factors tied to successful organizational transitions across business units
- Synthesize findings into a mock HRBP dashboard of key reorg metrics, presenting to SVP to help HRBPs track retention, mobility, and headcount during reorganizations
- Architect a 6-stage Python pipeline ingesting 1,459 Snowflake performance reviews through PII anonymization, BGE semantic filtering, and FAISS retrieval before routing to an LLM DWA scoring agent
- Build a custom LLM client with sliding-window token rate limiting and retry-aware backoff to maintain reliable multi-hour inference throughput across the production review-processing pipeline

**Data Science Union** Los Angeles, California

*External Vice President* *Dec 2025 – Present*

- Directed external strategy across client partnerships, marketing, and professional development, expanding consulting and career opportunities for members
- Managed 3 directors and coordinated executive board priorities, aligning cross-functional execution to deliver quarterly and annual objectives
- Cultivated industry and alumni partnerships, securing client data science projects and expanding career programming across the organization

**Kohl's** Los Angeles, California

*Project Manager and Data Analyst* *Jun 2025 – Mar 2026*

- Processed 8.5M+ logistics records in BigQuery with SQL, building validation logic that removed duplicate records and impossible event sequences
- Developed and deployed a two-stage XGBoost forecasting system simulating live milestone updates, cutting shipment arrival MAE by 83%

- Engineered feature pipelines for route, carrier, and seasonality signals in Vertex AI Workbench, improving transit time forecast accuracy

**ACTUM Digital** Prague, Czech Republic

*Data Analyst and AI Intern* *Jun 2025 – Aug 2025*

- Built agentic Python pipelines to filter documents, generate metadata, and embed content into a Milvus vector database for AI-assisted retrieval
- Integrated an MCP server and authored prompt workflows for an internal AI assistant, co-presenting the live demo to ACTUM executives
- Automated extraction of structured data from 600+ unstructured PDFs with Python scripts, consolidating output into a centralized reporting dataset

**Million Dollar Baby** Los Angeles, California

*Data Analysis Consultant* *Apr 2025 – Jun 2025*

- Analyzed $10M+ in advertising spend with SQL, surfacing efficiency insights that informed growth strategy and marketing investment decisions
- Presented CAC and RoAS assessments across multiple attribution windows, guiding stakeholder decisions on marketing spend allocation
- Built an Excel dashboard tracking 20+ marketing KPIs, giving brand managers a recurring view of acquisition and return trends

## Technical Skills

**Languages**: Python, C++, Java, SQL, R, JavaScript
**Libraries/Frameworks**: PyTorch, scikit-learn, pandas, NumPy, Matplotlib, TensorFlow, Keras, XGBoost, LangChain, MLflow
**Tools & Platforms**: Git, GitHub, AWS, GCP, Claude Code, BigQuery, Snowflake, Vertex AI, Tableau, Jupyter Notebooks